

On a proof of the Gorenstein Symmetry Conjecture

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Abstract

We give a homological proof of the famous Gorenstein Symmetry Conjecture, which asserts that one-sided finiteness of the self-injective dimension of an Artin algebra is sufficient for Gorensteinness.

Key words: Gorenstein Symmetry Conjecture; Artinian rings; Cotorsion theory

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1 Introduction

Throughout this paper, Λ is an associative ring with identity and all Λ -modules are unitary right Λ -modules. We view left Λ -modules as right ones over the opposite ring Λ^{op} . We denote by $(\text{mod-}\Lambda)$ $\text{Mod-}\Lambda$ the class of all (finitely generated) right Λ -modules. The symbols $\text{id}_{\Lambda}(M)$ and $\text{pd}_{\Lambda}(M)$ stand for the injective and projective dimension of a right Λ -module M , respectively.

The main purpose of this paper is to give a positive answer to the following long-standing conjecture posed by Auslander and Reiten in [2, p. 150] (see also [3, Conjecture (13)]).

Gorenstein Symmetry Conjecture (GSC) *Let Λ be an Artin algebra with $\text{id}_{\Lambda}(\Lambda) < \infty$. Then $\text{id}_{\Lambda^{\text{op}}}(\Lambda^{\text{op}}) < \infty$.*

For an Artin algebra Λ , we learn from [2, p. 121] and [12, Corollary 2] that $\text{id}_{\Lambda}(\Lambda) \leq 1$ if and only if $\text{id}_{\Lambda^{\text{op}}}(\Lambda^{\text{op}}) \leq 1$. Later, Beligiannis [4, Theorem 11.4] proved that GSC holds for the class of virtually Gorenstein Artin algebras.

We show in Theorem 3.9 that GSC is true for any Artin algebra.

We continue the introduction by recalling some basic definitions and notions that we use in this paper. Given a class $\mathcal{C}$ of Λ -modules, we denote

- ${}^{\perp}\mathcal{C} = \{X : \text{Ext}_{\Lambda}^1(X, C) = 0 \text{ for all } C \in \mathcal{C}\},$
- $\mathcal{C}^{\perp} = \{X : \text{Ext}_{\Lambda}^1(C, X) = 0 \text{ for all } C \in \mathcal{C}\},$
- ${}^{\perp\infty}\mathcal{C} = \{X : \text{Ext}_{\Lambda}^i(X, C) = 0 \text{ for all } C \in \mathcal{C} \text{ and } i \geq 1\},$
- $\mathcal{C}^{\perp\infty} = \{X : \text{Ext}_{\Lambda}^i(C, X) = 0 \text{ for all } C \in \mathcal{C} \text{ and } i \geq 1\},$
- $\text{Proj}(\Lambda)$ = the class of all projective right Λ -modules,
- $\text{Inj}(\Lambda)$ = the class of all injective right Λ -modules.

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(Pre)covers and (pre)envelopes. Let $\mathcal{C}$ be a class of modules in $\text{Mod-}\Lambda$. A morphism $\phi : C \rightarrow M$ of $\text{Mod-}\Lambda$ with $C \in \mathcal{C}$ is called a $\mathcal{C}$ -precover [7] of M if for any morphism $f : C' \rightarrow M$ with $C' \in \mathcal{C}$, there is a morphism $g : C' \rightarrow C$ such that $\phi g = f$. Moreover, if the only such g are automorphisms of C when $C' = C$ and $f = \phi$, then the $\mathcal{C}$ -precover ϕ is called a $\mathcal{C}$ -cover. Dually, we have the definitions of a $\mathcal{C}$ -preenvelope and a $\mathcal{C}$ -envelope. We say that $\mathcal{C}$ is *(pre)covering* (*(pre)enveloping*) in case every module in $\text{Mod-}\Lambda$ has a $\mathcal{C}$ -(pre)cover ($\mathcal{C}$ -(pre)envelope).

A module M is said to have a *special $\mathcal{C}$ -precover*, if there is an exact sequence $0 \rightarrow K \rightarrow C \rightarrow M \rightarrow 0$ with $C \in \mathcal{C}$ and $K \in \mathcal{C}^\perp$. M is said to have a *special $\mathcal{C}$ -preenvelope*, if there is an exact sequence $0 \rightarrow M \rightarrow C \rightarrow L \rightarrow 0$ with $C \in \mathcal{C}$ and $L \in {}^\perp\mathcal{C}$.

Cotorsion theory. A pair $(\mathcal{X}, \mathcal{Y})$ of subcategories of $\text{Mod-}\Lambda$ is said to be a *cotorsion theory* [7] if $\mathcal{X}^\perp = \mathcal{Y}$ and ${}^\perp\mathcal{Y} = \mathcal{X}$. A cotorsion theory $(\mathcal{X}, \mathcal{Y})$ is called *complete* if every object in $\text{Mod-}\Lambda$ has a special $\mathcal{X}$ -precover and a special $\mathcal{Y}$ -preenvelope. A cotorsion theory $(\mathcal{X}, \mathcal{Y})$ is said to be *hereditary* if whenever $0 \rightarrow Y \rightarrow Y' \rightarrow Y'' \rightarrow 0$ is exact with $Y, Y' \in \mathcal{Y}$ then $Y'' \in \mathcal{Y}$.

2 A characterization of modules of virtually finite injective dimension

In this section, we give a characterization of *modules of virtually finite injective dimension* [4] over right Noetherian rings.

First, we recall the definitions of Gorenstein projective and injective modules introduced by Enochs and Jenda in [8].

Definition 2.1. A *complete projective resolution* is an exact sequence of projective Λ -modules,

$$\cdots \rightarrow P_1 \rightarrow P_0 \rightarrow P_{-1} \rightarrow P_{-2} \rightarrow \cdots,$$

such that $\text{Hom}_\Lambda(-, Q)$ leaves the sequence exact whenever Q is a projective Λ -module; the module $M = \text{im}(P_0 \rightarrow P_{-1})$ is then said to be *Gorenstein projective*.

Gorenstein injective modules are defined dually.

We use $\mathcal{GP}$ and $\mathcal{GI}$ to denote, respectively, the classes of all Gorenstein projective and injective (right) Λ -modules. The modules in ${}^\perp\mathcal{GI}$ are called *modules of virtually finite injective dimension* [4].

We need the following proposition due to Šaroch and Šťovíček.

Proposition 2.2. ([18, Theorem 5.6]). *For any ring Λ , $({}^\perp\mathcal{GI}, \mathcal{GI})$ is a hereditary complete cotorsion theory.*

We recall some definitions from [17] and [7].

Definition 2.3. Let μ be an ordinal and $\mathcal{A} = (A_\alpha \mid \alpha \leq \mu)$ be a sequence of modules. If $A_0 = 0$, $A_\alpha \subseteq A_{\alpha+1}$ for all $\alpha < \mu$ and $A_\alpha = \bigcup_{\beta < \alpha} A_\beta$ for all limit ordinals $\alpha \leq \mu$, then the sequence $\mathcal{A}$ is called a *continuous (well ordered) chain of modules*.

Let M be a module and $\mathcal{X}$ be a class of modules. M is $\mathcal{X}$ -filtered, provided that there is an ordinal μ and a continuous chain of modules, $(M_\alpha \mid \alpha \leq \mu)$, consisting of submodules of M such that $M = M_\mu$, and each of the modules $M_{\alpha+1}/M_\alpha$ ($\alpha < \mu$) is isomorphic to an element of $\mathcal{X}$. Then chain $(M_\alpha \mid \alpha \leq \mu)$ is called an $\mathcal{X}$ -filtration of M . If μ is finite, then M is said to be *finitely $\mathcal{X}$ -filtered*.

For convenience, we list the following classical result from [17].

Lemma 2.4. ([17, Corollary 6.14]). *Let $\mathcal{X}$ be a set of modules containing Λ . Then the class ${}^\perp(\mathcal{X}^\perp)$ consists of all direct summands of $\mathcal{X}$ -filtered modules.*

We get a useful corollary.

Corollary 2.5. *Let $\mathcal{X}$ be a set of modules containing Λ . Then any simple module in ${}^\perp(\mathcal{X}^\perp)$ is isomorphic to a direct summand of some module in $\mathcal{X}$.*

Proof. Let S be a simple module in ${}^\perp(\mathcal{X}^\perp)$. Then S is isomorphic to a direct summand of M which is $\mathcal{X}$ -filtered by Lemma 2.4. There is an ordinal μ and a continuous (well ordered) chain $(M_\alpha \mid \alpha \leq \mu)$ of submodules of M such that $M = M_\mu$, and each of the modules $M_{\alpha+1}/M_\alpha$ ($\alpha < \mu$) is isomorphic to an element of $\mathcal{X}$. We may assume that S is a submodule of M . Then the subset

$$\kappa = \{\alpha \leq \mu \mid S \subseteq M_\alpha\}$$

is nonempty. Note that μ is well ordered. Hence κ has a least element (cf. [7, Definition 1.1.4]), say β . If β is a limit ordinal, then $M_\beta = \bigcup_{\gamma < \beta} M_\gamma$. But S is simple. Thus $S \subseteq M_\gamma$ for some $\gamma < \beta$, in contradiction to minimality. Thus β is not a limit ordinal. Since $M_0 = 0$, we have that $\beta \neq 0$. Therefore β is a successor ordinal, i.e., $\beta = \lambda + 1$ for some $\lambda < \mu$. Then S is not contained in M_λ , and hence $S \cap M_\lambda = \{0\}$ since S is simple. Note that S is a direct summand of M . So S is also a direct summand of $M_{\lambda+1}$. It follows that S is isomorphic to a direct summand of $M_{\lambda+1}/M_\lambda$ which is isomorphic to an element of $\mathcal{X}$. This finishes the proof. $\square$

The following homological lemma is a generalization of [7, Theorem 7.3.4].

Lemma 2.6. ([15, Lemma 3.8].) *Let M and N be Λ -modules, n a positive integer and M the union of a continuous chain of submodules $(M_\alpha \mid \alpha \leq \kappa)$. If $\text{Ext}_\Lambda^n(M_0, N) = 0$ and $\text{Ext}_\Lambda^n(M_{\alpha+1}/M_\alpha, N) = 0$ whenever $\alpha + 1 < \kappa$ then $\text{Ext}_\Lambda^n(M, N) = 0$*

To simplify the proof of the next theorem, we give the following notation.

Notation 2.7. Let Λ be a right Noetherian ring. Then there is a family $(E_j)_{j \in J}$ of indecomposable injective right Λ -modules such that every injective right Λ -module is the direct sum of copies of the various E_j (see the proof of [7, Theorem 5.4.1]). For each E_j , take a projective resolution

$$\cdots \xrightarrow{f_{j2}} F_{j1} \xrightarrow{f_{j1}} F_{j0} \xrightarrow{f_{j0}} E_j \longrightarrow 0 \quad (\spadesuit_j)$$

of E_j , and let

$$0 \longrightarrow \Lambda_\Lambda \xrightarrow{h_0} E_0 \xrightarrow{h_1} E_1 \xrightarrow{h_2} \cdots \quad (\clubsuit)$$

be a minimal injective resolution of Λ_Λ . Put $\mathcal{X} = \{\text{im}(f_{ji}), \text{im}(h_i) \mid i \geq 0, j \in J\}$.

Now we give a characterization of modules of virtually finite injective dimension over right Noetherian rings.

Theorem 2.8. *Let Λ be a right Noetherian ring and keep the notation as above. Then $({}^\perp(\mathcal{X}^\perp), \mathcal{X}^\perp)$ is a hereditary complete cotorsion theory. Moreover, ${}^\perp\mathcal{G}\mathcal{J} = {}^\perp(\mathcal{X}^\perp)$, i.e., $\mathcal{X}^\perp = \mathcal{G}\mathcal{J}$.*

Proof. Since $\mathcal{X}$ is a set, $({}^\perp(\mathcal{X}^\perp), \mathcal{X}^\perp)$ is a complete cotorsion theory by [7, Theorem 7.4.1]. Next we prove that it is hereditary.

Since each $\text{im}(f_{ji})$ belongs to ${}^\perp(\mathcal{X}^\perp)$, for any $n \geq 1$, we deduce from the exact sequence $(\spadesuit_j)$ that $\text{Ext}_\Lambda^n(\text{im}(f_{ji}), Y) = 0$ for all $Y \in \mathcal{X}^\perp$. Therefore, $\text{im}(f_{ji}) \in {}^{\perp\infty}(\mathcal{X}^\perp)$. On the other hand, every injective right Λ -module is the direct sum of copies of the various E_j . Hence $\mathbf{Inj}(\Lambda) \subseteq {}^{\perp\infty}(\mathcal{X}^\perp)$. But then we see from the exact sequence $(\clubsuit)$ that $\text{im}(h_i) \in {}^{\perp\infty}(\mathcal{X}^\perp)$. In addition, the class ${}^\perp(\mathcal{X}^\perp)$ consists of all direct summands of $\mathcal{X}$ -filtered modules by Lemma 2.4. These observations together with Lemma 2.6 tell us that, $\text{Ext}_\Lambda^n(B, Y) = 0$ for all $B \in {}^\perp(\mathcal{X}^\perp)$, $Y \in \mathcal{X}^\perp$, and $n \geq 1$. So the pair $({}^\perp(\mathcal{X}^\perp), \mathcal{X}^\perp)$ is hereditary by [17, Lemma 5.24].

Now we prove that $\mathcal{X}^\perp = \mathcal{G}\mathcal{I}$. Since injective modules are contained in the thick class ${}^\perp\mathcal{G}\mathcal{I}$ (cf. [18, Lemma 5.24]), we see from the exact sequences $(\spadesuit_j)$ and $(\clubsuit)$ that $\mathcal{X} \subseteq {}^\perp\mathcal{G}\mathcal{I}$. Hence $\mathcal{X}^\perp \supseteq ({}^\perp\mathcal{G}\mathcal{I})^\perp = \mathcal{G}\mathcal{I}$ (cf. Proposition 2.2).

It remains to show that $\mathcal{X}^\perp \subseteq \mathcal{G}\mathcal{I}$. For any $M \in \mathcal{X}^\perp$, there is an exact sequence $0 \rightarrow K \rightarrow \Lambda^{(\lambda)} \rightarrow M \rightarrow 0$ for some set λ . Then the sequence $0 \rightarrow \Lambda^{(\lambda)} \rightarrow E_0^{(\lambda)} \rightarrow (\text{im}(h_1))^{(\lambda)} \rightarrow 0$ is also exact. Construct a pushout of $\Lambda^{(\lambda)} \rightarrow E_0^{(\lambda)}$ along $\Lambda^{(\lambda)} \rightarrow M$, we obtain a commutative diagram with exact rows and columns:

$$\begin{array}{ccccccc}
& & 0 & & 0 & & \\
& & \downarrow & & \downarrow & & \\
0 & \longrightarrow & K & \longrightarrow & \Lambda^{(\lambda)} & \longrightarrow & M \longrightarrow 0 \\
& & \parallel & & \downarrow & & \downarrow \\
0 & \longrightarrow & K & \longrightarrow & E_0^{(\lambda)} & \longrightarrow & Q \longrightarrow 0 \\
& & & & \downarrow & & \downarrow \\
& & & & \text{im}(h_1)^{(\lambda)} & \xlongequal{\quad} & \text{im}(h_1)^{(\lambda)} \\
& & & & \downarrow & & \downarrow \\
& & & & 0 & & 0
\end{array}$$

Note that $\text{Ext}_\Lambda^1(\text{im}(h_1)^{(\lambda)}, M) \cong (\text{Ext}_\Lambda^1(\text{im}(h_1), M))^\lambda = 0$. Thus $Q \cong M \oplus \text{im}(h_1)^{(\lambda)}$, and we get an epimorphism $E_0^{(\lambda)} \rightarrow M$ with $E_0^{(\lambda)}$ injective (Λ is right Noetherian). Hence there exists a short exact sequence

$$0 \longrightarrow K \longrightarrow I_0 \xrightarrow{g_0} M \longrightarrow 0$$

where g_0 is an injective cover. Then $K \in \mathbf{Inj}(\Lambda)^\perp$ by [7, Corollary 7.2.3]. But $M \in \mathbf{Inj}(\Lambda)^{\perp\infty}$. Thus, one can see from the above short exact sequence that $K \in \mathbf{Inj}(\Lambda)^{\perp\infty}$. For any $i \geq 1$, we have an exact sequence

$$0 = \text{Ext}_\Lambda^1(\text{im}(h_i), M) \rightarrow \text{Ext}_\Lambda^2(\text{im}(h_i), K) \rightarrow \text{Ext}_\Lambda^2(\text{im}(h_i), I_0) = 0.$$

So $\text{Ext}_\Lambda^2(\text{im}(h_i), K) = 0$. Now consider the short exact sequence $0 \rightarrow \text{im}(h_{i-1}) \rightarrow E_{i-1} \rightarrow \text{im}(h_i) \rightarrow 0$ which induces an exact sequence

$$0 = \text{Ext}_\Lambda^1(E_{i-1}, K) \rightarrow \text{Ext}_\Lambda^1(\text{im}(h_{i-1}), K) \rightarrow \text{Ext}_\Lambda^2(\text{im}(h_i), K) = 0.$$

Thus $\text{Ext}_\Lambda^1(\text{im}(h_{i-1}), K) = 0$ for any $i \geq 1$. This implies $K \in \mathcal{X}^\perp$. Recursively, we construct an exact sequence of right Λ -modules

$$\dots \xrightarrow{g_2} I_1 \xrightarrow{g_1} I_0 \xrightarrow{g_0} M \longrightarrow 0$$

with $I_i \in \mathbf{Inj}(\Lambda)$ and $\ker(g_i) \in \mathbf{Inj}(\Lambda)^{\perp_\infty}$ ($i \geq 0$). Therefore, $M \in \mathcal{GS}$ by [10, Lemma 17(2)], and $\mathcal{X}^\perp \subseteq \mathcal{GS}$, as required. $\square$

3 Proof of the Gorenstein Symmetry Conjecture for artin algebras

We begin with the following definition.

Definition 3.1. Let M be a Λ -module. We say that M has *Gorenstein projective dimension* at most n , and we write $\mathrm{Gpd}_\Lambda(M) \leq n$, if there exists an exact sequence of Λ -modules $0 \rightarrow G_n \rightarrow \cdots \rightarrow G_0 \rightarrow M \rightarrow 0$ where each G_i is Gorenstein projective. If there is no such n , set $\mathrm{Gpd}_\Lambda(M) = \infty$.

The *Gorenstein injective dimension*, $\mathrm{Gid}_\Lambda(M)$, is defined similarly.

Bennis and Mahdou [6, Theorem 1.1] proved the following equality

$$\sup\{\mathrm{Gpd}_\Lambda(M) \mid M \text{ is a right } \Lambda\text{-module}\} = \sup\{\mathrm{Gid}_\Lambda(M) \mid M \text{ is a right } \Lambda\text{-module}\}.$$

The common value of the above quantities is called the *right Gorenstein global dimension* of Λ and we denote it by $r.\mathrm{Ggldim}(\Lambda)$.

It is known that $r.\mathrm{Ggldim}(\Lambda)$ is related to the cohomological invariants $r.\mathrm{silp}(\Lambda) := \sup\{\mathrm{id}_\Lambda(P) \mid P \in \mathbf{Proj}(\Lambda)\}$ and $r.\mathrm{spli}(\Lambda) := \sup\{\mathrm{pd}_\Lambda(E) \mid E \in \mathbf{Inj}(\Lambda)\}$ introduced by Gedrich and Gruenberg in [11] via the formula

$$r.\mathrm{Ggldim}(\Lambda) = \max\{r.\mathrm{silp}(\Lambda), r.\mathrm{spli}(\Lambda)\} \quad ([5, \text{Theorem 3.3}] \text{ and } [16, \text{Theorem 2.1}]).$$

In addition, if $r.\mathrm{Ggldim}(\Lambda) < \infty$, then $r.\mathrm{Ggldim}(\Lambda) = r.\mathrm{silp}(\Lambda) = r.\mathrm{spli}(\Lambda)$ by [16, Proposition 2.3].

As usual, we call a ring Λ Noetherian in case it is both left Noetherian and right Noetherian.

Proposition 3.2. *Let Λ be a Noetherian ring. Then*

$$r.\mathrm{Ggldim}(\Lambda) = r.\mathrm{Ggldim}(\Lambda^{\mathrm{op}}).$$

Proof. This follows from [14, Corollary 2.7]; see also [5, Theorem 3.10]. $\square$

Recall that an epimorphism $f : A \twoheadrightarrow B$ of Λ -modules is said to be *pure* if the map $f^* : \mathrm{Hom}_\Lambda(F, A) \rightarrow \mathrm{Hom}_\Lambda(F, B)$ is epic for all finitely presented Λ -modules F ; then $\ker(f)$ is said to be *pure* in A .

Proposition 3.3. *Let Λ be a right Noetherian ring, κ an infinite cardinal with $\mathrm{Card}(\Lambda) \leq \kappa$. Let $\mathcal{GS}_\kappa$ denote an irredundant set of representatives of modules G such that $G \in \mathcal{GS}$ and $\mathrm{Card}(G) \leq \kappa$. Assume that the class $\mathcal{GS}$ is closed under pure submodules. Then any $M \in \mathcal{GS}$ is $\mathcal{GS}_\kappa$ -filtered.*

Proof. First, we prove that $\mathcal{GS}$ is closed under direct limits.

Let $\{G_i\}_{i \in I}$ be any direct system of modules from $\mathcal{GS}$. Then we get a pure exact sequence $0 \rightarrow A \rightarrow \bigoplus G_i \rightarrow \varinjlim G_i \rightarrow 0$ by [17, Corollary 2.9]. Note that $\bigoplus G_i$ is pure in $\prod G_i$ by [17, Corollary 2.21(c)]. So both $\bigoplus G_i$ and A belong to $\mathcal{GS}$. It follows that $\varinjlim M_i \in \mathcal{GS}$ by Proposition 2.2.

Let $M \in \mathcal{GS}$. We mimic the proof of [9, Corollary 3.3.4] to show that M is $\mathcal{GS}_\kappa$ -filtered. For any $x \in M$, by [7, Lemma 5.3.12], there is a pure submodule M_0 of M with $x \in M_0$ such that $\mathrm{Card}(M_0) \leq \kappa$ (simply

let $N = x\Lambda$ and f the inclusion map from N to M in the lemma). We see that both M_0 and M/M_0 are in $\mathcal{G}\mathcal{S}$ by Proposition 2.2.

For any $x_1 + M_0 \in M/M_0$, again by [7, Lemma 5.3.12], there is a pure submodule M_1/M_0 of M/M_0 such that $x_1 + M_0 \in M_1/M_0$ and $\text{Card}(M_1/M_0) \leq \kappa$. But $M_1/M_0 \in \mathcal{G}\mathcal{S}$, so it is isomorphic to an element of $\mathcal{G}\mathcal{S}_\kappa$. Note that $\mathcal{G}\mathcal{S}$ is closed under extensions. Hence $M_1 \in \mathcal{G}\mathcal{S}$. Thus $M/M_1 \in \mathcal{G}\mathcal{S}$ again by Proposition 2.2.

Proceeding by induction, we get a chain of modules $M_n \in \mathcal{G}\mathcal{S}$, $n \in \mathbb{N}$, such that M_{n+1}/M_n is isomorphic to an element of $\mathcal{G}\mathcal{S}_\kappa$. Let $M_{\omega_0} = \bigcup_{n \in \mathbb{N}} M_n$, then $M_{\omega_0} \in \mathcal{G}\mathcal{S}$ and $M/M_{\omega_0} \in \mathcal{G}\mathcal{S}$. Choosing $x_{\omega_0+1} + M_{\omega_0} \in M/M_{\omega_0}$, we find a pure submodule $M_{\omega_0+1}/M_{\omega_0}$ of M/M_{ω_0} such that $x_{\omega_0+1} + M_{\omega_0} \in M_{\omega_0+1}/M_{\omega_0}$, $\text{Card}(M_{\omega_0+1}/M_{\omega_0}) \leq \kappa$ and $M_{\omega_0+1}/M_{\omega_0} \in \mathcal{G}\mathcal{S}$. So $M_{\omega_0+1}/M_{\omega_0}$ is isomorphic to an element of $\mathcal{G}\mathcal{S}_\kappa$.

We have shown that $\mathcal{G}\mathcal{S}$ is closed under direct limits. Hence, by transfinite induction we can write M as a union of a continuous chain $(M_\alpha)_{\alpha < \lambda}$ of pure submodules of M , such that $M_0 \in \mathcal{G}\mathcal{S}_\kappa$, $M_{\alpha+1}/M_\alpha$ is isomorphic to an element of $\mathcal{G}\mathcal{S}_\kappa$ whenever $\alpha + 1 < \lambda$, i.e., M is $\mathcal{G}\mathcal{S}_\kappa$ -filtered. The proof is finished. $\square$

To simplify the proof of the next theorem, we give the following notation.

Notation 3.4. Let Λ be a right Noetherian ring and $(E_j)_{j \in J}$ the family be as in Notation 2.7. Let κ be an infinite cardinal with $\text{Card}(\kappa) \geq \max\{\text{Card}(\Lambda), \text{Card}(\bigoplus_{j \in J} E_j)\}$. Let $\mathcal{U}$ denote an irredundant set of representatives of right Λ -modules A such that $\text{Gid}_\Lambda(A) < \infty$ and $\text{Card}(A) \leq \kappa$. For each $A_u \in \mathcal{U}$, choose a projective resolution

$$\cdots \xrightarrow{\pi_{u2}} P_{u1} \xrightarrow{\pi_{u1}} P_{u0} \xrightarrow{\pi_{u0}} A_u \longrightarrow 0 \quad (\heartsuit_u)$$

of A_u , and let

$$0 \longrightarrow A_u \xrightarrow{\tau_{u0}} I_{u0} \xrightarrow{\tau_{u1}} I_{u1} \xrightarrow{\tau_{u2}} \cdots \quad (\sharp_u)$$

be a minimal injective resolution of A_u . Put $\overline{\mathcal{U}} = \{\text{im}(\pi_{ui}), \text{im}(\tau_{ui}) \mid i \geq 0, u \in \mathcal{U}\}$.

We believe that the following result is of interest.

Theorem 3.5. *Let Λ be a right Noetherian ring and keep the notation as above. If $\text{Gid}_\Lambda(\Lambda) < \infty$ and the class $\mathcal{G}\mathcal{S}$ is closed under pure submodules, then:*

- (1) *every right Λ -module M is isomorphic to a direct summand a $\overline{\mathcal{U}}$ -filtered module;*
- (2) *$\text{Gid}_\Lambda(S) < \infty$ for any simple module S .*

Proof. (1) Since $\text{Gid}_\Lambda(\Lambda) < \infty$ and $\text{Card}(\kappa) \geq \max\{\text{Card}(\Lambda), \text{Card}(\bigoplus_{j \in J} E_j)\}$, we may assume that both Λ and $\bigoplus_{j \in J} E_j$ belong to the set $\mathcal{U}$. Hence we may assume that the set $\mathcal{X}$ constructed in Notation 2.7 is contained in $\overline{\mathcal{U}}$. Therefore, $\overline{\mathcal{U}}^\perp \subseteq \mathcal{X}^\perp = \mathcal{G}\mathcal{S}$ (cf. Theorem 2.8), and $({}^\perp(\overline{\mathcal{U}}^\perp), \overline{\mathcal{U}}^\perp)$ is a complete cotorsion theory by [7, Theorem 7.4.1]. Next we prove that the pair $({}^\perp(\overline{\mathcal{U}}^\perp), \overline{\mathcal{U}}^\perp)$ is hereditary.

Since each $\text{im}(\pi_{ui})$ belongs to ${}^\perp(\overline{\mathcal{U}}^\perp)$, for any $n \geq 1$, we deduce from the exact sequence $(\heartsuit_u)$ that $\text{Ext}_\Lambda^n(\text{im}(\pi_{ui}), Y) = 0$ for all $Y \in \overline{\mathcal{U}}^\perp$. Therefore, $\text{im}(\pi_{ui}) \in {}^{\perp_\infty}(\overline{\mathcal{U}}^\perp)$. On the other hand, every injective right Λ -module is the direct sum of copies of the various E_j . Hence $\text{Inj}(\Lambda) \subseteq {}^{\perp_\infty}(\overline{\mathcal{U}}^\perp)$. But then we see from the exact sequence $(\sharp_u)$ that $\text{im}(\tau_{ui}) \in {}^{\perp_\infty}(\overline{\mathcal{U}}^\perp)$. In addition, the class ${}^\perp(\overline{\mathcal{U}}^\perp)$ consists of all direct summands of $\overline{\mathcal{U}}$ -filtered modules by Lemma 2.4. These observations together with Lemma 2.6 tell us that, $\text{Ext}_\Lambda^n(W, Y) = 0$ for all $W \in {}^\perp(\overline{\mathcal{U}}^\perp)$, $Y \in \overline{\mathcal{U}}^\perp$, and $n \geq 1$. So the pair $({}^\perp(\overline{\mathcal{U}}^\perp), \overline{\mathcal{U}}^\perp)$ is hereditary by [17, Lemma 5.24].

For any $M \in \text{Mod-}\Lambda$, there is an exact sequence $0 \rightarrow M \rightarrow Q \rightarrow L \rightarrow 0$ with $Q \in \overline{\mathcal{U}}^\perp \subseteq \mathcal{G}\mathcal{S}$ and $L \in {}^\perp(\overline{\mathcal{U}}^\perp)$. Note that the class $\mathcal{G}\mathcal{S}$ is closed under pure submodules. So Q is $\mathcal{G}\mathcal{S}_\kappa$ -filtered by Proposition

3.3. By construction, every module in $\mathcal{G}\mathcal{S}_\kappa$ is isomorphic to an element of $\overline{\mathcal{U}}$. Hence Q is $\overline{\mathcal{U}}$ -filtered. This implies $Q \in {}^\perp(\overline{\mathcal{U}}^\perp)$. It follows that $M \in {}^\perp(\overline{\mathcal{U}}^\perp)$. This proves (1).

Now let $S \in \text{Mod-}\Lambda$ be simple. Since $S \in {}^\perp(\overline{\mathcal{U}}^\perp)$, S is isomorphic to a direct summand of some module Y in $\overline{\mathcal{U}}$ by Corollary 2.5. According to the construction of $\overline{\mathcal{U}}$, $\text{Gid}_\Lambda(Y) < \infty$. Then the dual of [13, Proposition 2.19] forces that $\text{Gid}_\Lambda(S) < \infty$. This finishes the proof. $\square$

The next lemma is well-known, but we could not find a concrete reference to it.

Lemma 3.6. *Let $0 \rightarrow B \rightarrow C \rightarrow D \rightarrow 0$ be a short exact sequence in $\text{Mod-}\Lambda$. If $\text{Gid}_\Lambda(B) \leq n < \infty$ and $\text{Gid}_\Lambda(D) \leq n$, then $\text{Gid}_\Lambda(C) \leq n$.*

Proof. This is a direct consequence of [13, Theorems 2.22 and 2.24]. $\square$

We also need the following result.

Lemma 3.7. *Let $n \geq 0$ be given and let $\mathcal{G}\mathcal{S}_n$ denote the class of right Λ -modules M such that $\text{Gid}_\Lambda(M) \leq n$. If $\mathcal{G}\mathcal{S}$ is closed under direct limits, then $\mathcal{G}\mathcal{S}_n$ is also closed under direct limits.*

Proof. Let $\{M_i\}_{i \in I}$ be any direct system of right Λ -modules with each $M_i \in \mathcal{G}\mathcal{S}_n$. Take an injective resolution

$$0 \longrightarrow M_i \xrightarrow{\theta_{i0}} G_{i0} \xrightarrow{\theta_{i1}} G_{i1} \xrightarrow{\theta_{i2}} \cdots$$

for each M_i and let $L_{in} = \text{im}(\theta_{in})$. Then $L_{in} \in \mathcal{G}\mathcal{S}$ by [13, Theorem 2.22 (iv)]. Since $\mathcal{G}\mathcal{S}$ is closed under direct limits, we see from the exactness of the following complex

$$0 \rightarrow \varinjlim M_i \rightarrow \varinjlim G_{i0} \rightarrow \varinjlim G_{i1} \rightarrow \cdots \rightarrow \varinjlim L_{in} \rightarrow 0$$

that $\varinjlim M_i \in \mathcal{G}\mathcal{S}_n$, as required. $\square$

Now we turn our attention to artinian rings.

Theorem 3.8. *Let Λ be an Artinian ring. If the classes of all Gorenstein injective right and left Λ -modules are closed under pure submodules, then the following quantities are identical:*

- (1) $r.\text{Ggldim}(\Lambda)$;
- (2) $r.\text{Ggldim}(\Lambda^{\text{op}})$;
- (3) $\text{id}_\Lambda(\Lambda)$;
- (4) $\text{id}_{\Lambda^{\text{op}}}(\Lambda^{\text{op}})$;
- (5) $\text{Gid}_\Lambda(\Lambda)$;
- (6) $\text{Gid}_{\Lambda^{\text{op}}}(\Lambda^{\text{op}})$.

Proof. It is clear that (5) $\leq$ (3), and [5, Theorem 3.3] tells us that (3) $\leq$ (1). Next we prove that (1) $\leq$ (5). We may assume that $\text{Gid}_\Lambda(\Lambda) < \infty$. Since Λ is Artinian, there is an irredundant finite set $\mathcal{S}$ of representatives of the simple modules in $\text{Mod-}\Lambda$. Then $\text{Gid}_\Lambda(S) < \infty$ for any $S \in \mathcal{S}$ by Theorem 3.5(2). So $n = \sup\{\text{Gid}_\Lambda(S) \mid S \in \mathcal{S}\} < \infty$. In addition, it is well-known that any $M \in \text{mod-}\Lambda$ is finitely $\mathcal{S}$ -filtered (cf. [1, Proposition 11.1]). By induction, we get from Lemma 3.6 that $\text{Gid}_\Lambda(M) \leq n$. But every module is the direct limit of its finitely generated submodules, and $\mathcal{G}\mathcal{S}_n$ is closed under direct limits by Lemma 3.7. Thus $\text{Gid}_\Lambda(Y) \leq n$ for any $Y \in \text{Mod-}\Lambda$. It follows from [6, Theorem 1.1] that $r.\text{Ggldim}(\Lambda) \leq n$. We then deduce from [6, Lemma 2.1] that $r.\text{Ggldim}(\Lambda) = \text{id}_\Lambda(\Lambda) < \infty$. But then we get from the dual of [13, Proposition 2.27] that $\text{id}_\Lambda(\Lambda) = \text{Gid}_\Lambda(\Lambda)$. Hence (1) $\leq$ (5). Therefore, (1) = (3) = (5).

The proof of (2) = (4) = (6) is symmetric.

Finally, apply Proposition 3.2 to get that (1) = (2) = (3) = (4) = (5) = (6). $\square$

We are now in a position to show the validity of the Gorenstein Symmetry Conjecture for all Artin algebras.

Theorem 3.9. *Let Λ be an Artin algebra. Then the following quantities are identical:*

- (1) $r.\text{Ggldim}(\Lambda)$;
- (2) $r.\text{Ggldim}(\Lambda^{\text{op}})$;
- (3) $\text{id}_{\Lambda}(\Lambda)$;
- (4) $\text{id}_{\Lambda^{\text{op}}}(\Lambda^{\text{op}})$;
- (5) $\text{Gid}_{\Lambda}(\Lambda)$;
- (6) $\text{Gid}_{\Lambda^{\text{op}}}(\Lambda^{\text{op}})$.

Proof. Since Λ is an Artin algebra, the classes of all Gorenstein injective right and left Λ -modules are closed under pure submodules by [4, Proposition 3.8(iii)]. Now apply Theorem 3.8. $\square$

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